

# ASTRATRACK — PERFORMANCE AUDIT REPORT

Run ID: RUN\_1789091943 | Timestamp: 2026-09-11T01:59:03Z | SIH 2026

OVERALL EVALUATION VERDICT: FAIL

## 1. Mission Configuration & Pipeline

|             |               |          |                    |
|-------------|---------------|----------|--------------------|
| Scenario    | 01 — BASELINE | Detector | Classical CV (HSV) |
| Controller  | PID_CONTROL   | Filter   | Kalman             |
| Disturbance | NORMAL        | Duration | 2.63 s             |

## 2. Operational & Kinematic Metrics (Actual Measured Telemetry)

| Metric Name        | Measured Value | Units | Metric Name         | Measured Value | Units |
|--------------------|----------------|-------|---------------------|----------------|-------|
| Avg Tracking Error | 22.26          | px    | Max Tracking Error  | 178.1          | px    |
| RMS Tracking Error | 33.8           | px    | Lock Retention Rate | 99.4           | %     |
| Acquisition Time   | 0.03           | s     | Re-Acquisition Time | 0.0            | s     |
| Target Loss Count  | 0              | count | Recoveries Count    | 0              | count |
| System Rate        | 11.2           | FPS   | Pipeline Latency    | 77.99          | ms    |
| Detector Inference | 12.35          | ms    | Detection Conf      | 0.929          | 0-1   |

## 3. Quantitative Mission Criteria Evaluation

| Criterion                   | Target Threshold | Measured Actual | Audit Status |
|-----------------------------|------------------|-----------------|--------------|
| Lock Retention Rate         | >= 85.0%         | 99.4%           | PASSED       |
| RMS Tracking Error          | <= 15.0 px       | 33.80 px        | FAILED       |
| Maximum Tracking Error      | <= 40.0 px       | 178.10 px       | FAILED       |
| Processing Pipeline Latency | <= 35.0 ms       | 77.99 ms        | FAILED       |
| Mean Re-Acquisition Time    | <= 2.00 s        | 0.00 s          | PASSED       |

**Integrity Guarantee:** All displayed metrics are derived strictly from live simulation telemetry. No placeholder or synthesized figures were used in this audit.